

SIH25104 — Real-Time AI Voice Cloning / Impersonation Detection

Scoped Research Gaps, Objectives, Proposed Solution & System Architecture (hackathon-buildable version)

1. Why This Version Is Scoped Down

The original literature review and gap analysis were sound — the ten identified gaps (G1–G10) and nine objectives (O1–O9) are all genuinely supported by the 29 reviewed papers. The issue was framing: presenting all nine objectives as "the proposed work" reads as a multi-year research program, not a hackathon deliverable. This version keeps every gap as legitimate motivation, but separates what is actually built and demoed for SIH25104 from what is honestly labeled future work. That separation is what should be presented to judges — it is more credible than an unscoped claim, not less impressive.

2. Research Gaps — Targeted vs. Future Work

The comparative analysis from the original literature review is preserved as background context. The table below re-labels each gap by whether it is directly targeted by a buildable objective in this submission.

ID	Gap	Why it's realistic to target now	Status
G1	No open-set / cross-dataset generalization check in most demo-stage projects	ASVspoof2019 (train) + IndieFake (cross-test) are both public and small enough to run in the build window	Core — targeted
G2	Detectors output only a binary real/fake label, no usable risk signal	A calibrated score + 3-tier risk bucket is a thin, buildable layer on top of any classifier's softmax output	Core — targeted
G3	Indian-speaker / Indic-language representation is thin in most training pipelines	Reporting (not solving) the accuracy drop on IndieFake is honest and demonstrable in a few hours of evaluation	Core — targeted (reported, not solved)
G4	Few prototypes simulate even a simplified streaming/real-time flow	Short fixed-length window inference + rolling aggregation is a lightweight, demoable approximation	Core — targeted (simplified)
G5	Full live-call stability, endpointing and streaming-system engineering	Requires production audio infrastructure and multi-week tuning	Future work
G6	Telephony/VoIP codec and compression robustness	Needs a curated degraded-channel test set and dedicated evaluation cycle	Future work
G7	Privacy-preserving, content-decoupled representations (SafeEar-style)	A genuinely separate research track requiring its own model and validation	Future work
G8	Broad multi-Indic-language coverage and subgroup fairness reporting	Needs multiple annotated regional datasets beyond what's available in the build window	Future work
G9	Neural-audio-codec-specific synthesis detection	Emerging attack family with limited public data as of this literature review	Future work
G10	Multimodal (audio + video) verification	Out of scope for an audio-only detection pipeline	Future work

ID	Gap	Why it's realistic to target now	Status
G11	PS asks for REST/gRPC APIs and SDKs for banking, contact-center, communication and telecom integration	A REST API + OpenAPI-generated SDKs + one mock integration demo is buildable; live production hookups are not	Core — targeted (REST only; gRPC + real hookups deferred)

3. PS Requirement Coverage Audit

The full problem-statement text (background, problem statement, proposed solution, key components, expected outcomes) was checked line by line against the buildable scope. Every PS bullet is accounted for below — either built (Core), built in a reduced/lite form, or explicitly named as future work. Nothing in the PS description is silently dropped; where something can't be built in the hackathon window, that's stated rather than implied.

PS component (verbatim ask)	How it's addressed	Status	Reference
Acoustic/spectral analysis for synthesis artifacts	Wav2Vec2 + LFCC front-end feeding the bonafide/spoof classifier	Core	O1
Prosody and behavioral analysis (rhythm, pitch, pauses, microvariations)	Lightweight prosodic statistics (pitch contour variance, pause timing) added alongside spectral embeddings — not a full behavioral-biometrics model	Core (lite)	O8
Cross-session consistency checks vs. historical genuine samples	Pretrained speaker-embedding cosine similarity against one enrolled reference sample per protected caller — not a trained speaker-ID system with a real historical archive	Core (lite)	O9
Continuous real-time risk-score computation	Simulated via short fixed-length window inference + rolling aggregation	Core (simplified)	O3, O4
Threshold-based alerting configurable per scenario (e.g. high-value transfers)	Fixed Low/Med/High demo thresholds; per-scenario configurability not built	Core (lite)	O4
Contextual enrichment using call origin, transaction context, fraud history	No live banking/CRM data source exists to enrich from in a hackathon build	Future work	FO8
Multi-channel alerts (UI, SMS/email, in-app)	UI prompt only in the demo; actual SMS/email dispatch not wired up	Core (lite) / Future work	O11 / FO10
Pre-transaction warning + secondary-verification recommendation	Risk tier mapped to a suggested action (call-back, MFA, escalate) shown in the UI	Core	O11
Configurable workflows for banks/enterprises/govt to define automated responses	Needs an admin-configuration engine and per-org rule storage — real product scope, not a demo feature	Future work	FO9
Minimal retention / on-device or edge inference	Raw audio buffer discarded immediately after inference; only features/scores are logged. True on-device/edge deployment (vs. this demo's server-side inference) is deferred	Core (lite)	O10

PS component (verbatim ask)	How it's addressed	Status	Reference
Anonymization / feature-only logging	Logs store extracted features and risk scores, never raw audio or transcripts	Core	O10
REST/gRPC APIs and SDKs for banking, contact-center, comms, telecom integration	REST (FastAPI) + OpenAPI-generated SDK stubs + one mock webhook demo; gRPC and real production hookups deferred	Core (lite) / Future work	O7 / FO7
Multiple Indian languages and regional accents	Cross-tested (not trained for) on an Indian-speaker dataset to report the generalization gap honestly; broad multi-Indic coverage not built	Core (reported) / Future work	O2 / FO4
Expected Outcomes (fraud reduction, trust, national cybersecurity alignment)	Pitch-level vision statements — kept as motivation/narrative for judges, not presented as measured results of the working demo	Narrative only	—

4. Objectives

4.1 Core objectives (built for this submission)

ID	Core objective (built for the submission)	Addresses	Deliverable
O1	Train a bonafide-vs-spoof classifier on ASVspoof2019 LA using Wav2Vec2 embeddings with a lightweight classifier head	G1	Working model with EER/accuracy on the ASVspoof2019 eval split
O2	Cross-test the trained model on IndieFake without retraining, and report the accuracy/EER delta honestly	G1, G3	Generalization report (in-domain vs. out-of-domain numbers, stated plainly)
O3	Simulate real-time operation via short fixed-length window inference with rolling-average aggregation over an uploaded/recorded clip	G4	Demo that processes audio in near-real-time chunks, not full offline batch scoring
O4	Convert raw model probability into a calibrated 3-tier risk score (Low / Medium / High) with a numeric confidence value	G2	Risk-scoring module, not a bare binary label
O5	Surface simple rule-based explanation tags alongside the risk score (e.g. which acoustic cue class triggered the flag)	G2	Lightweight explainability layer — not a full interpretability framework
O6	Package the above into a demo interface (upload or record audio → risk report)	—	Working end-to-end demo for judges
O7	Expose the pipeline behind a documented REST API (FastAPI) with an OpenAPI spec, auto-generated client SDK stubs, and a mock webhook demo simulating a contact-center/call-platform integration	G11	Live API endpoint + generated SDKs + one working mock-integration example

ID	Core objective (built for the submission)	Addresses	Deliverable
O8	Add lightweight prosodic features (pitch-contour variance, pause/rhythm timing) alongside the spectral embeddings feeding the classifier	PS: prosody/behavioral analysis	Measurable lift (or documented non-lift) from adding prosody features vs. spectral-only
O9	Add a cross-session consistency signal: pretrained speaker-embedding cosine similarity between the incoming call and one enrolled reference sample per protected caller (e.g. a CXO)	PS: cross-session consistency checks	Speaker-similarity score fused into the final risk score
O10	Design and implement privacy-by-design handling: discard the raw audio buffer immediately after inference; persist only extracted features and risk scores in logs	PS: privacy & compliance module	No raw audio at rest after inference; a documented logging policy
O11	Add an Alert & Recommendation layer that maps the risk tier to a suggested action (call-back, MFA, escalate to supervisor) shown as a pre-transaction warning in the demo UI	PS: alerting & user interaction layer	Visible recommendation text tied to each risk tier in the demo

4.2 Future-work objectives (explicitly not claimed as built)

These remain valid research directions raised by the literature and should be presented as the project's roadmap, not its current state:

- FO1 (→ G5): Build a genuine streaming pipeline with endpointing and live-call stability testing.
- FO2 (→ G6): Evaluate and harden the model against telephony/VoIP codec and compression degradation.
- FO3 (→ G7): Investigate privacy-preserving, content-decoupled acoustic representations.
- FO4 (→ G8): Extend evaluation to broader Indic languages and report subgroup/fairness metrics.
- FO5 (→ G9): Add detection coverage for neural-audio-codec-synthesized speech.
- FO6 (→ G10): Explore multimodal audio+video verification for high-risk cases.
- FO7 (→ G11): Add a gRPC service alongside REST, and pursue real integrations with named core banking systems, telecom carrier APIs, and contact-center platforms (e.g. Genesys, Five9) — this needs vendor partnerships and compliance sign-off no hackathon team has in-window.
- FO8 (→ PS: contextual enrichment): Integrate live call-origin, transaction-context and fraud-history metadata from real banking/CRM systems — no such data source exists to connect to in a hackathon build.
- FO9 (→ PS: configurable workflows): Build an admin-configuration engine letting banks/enterprises/govt agencies define their own automated response rules per risk threshold.
- FO10 (→ PS: multi-channel alerts): Wire up actual SMS/email dispatch (e.g. via Twilio/SES) instead of an in-demo UI prompt only.
- FO11 (→ PS: privacy & compliance): Pursue formal compliance certification (DPDP/GDPR-style audit) and true on-device/edge inference deployment, beyond this demo's discard-after-inference policy.

5. Proposed Solution

A real-time-simulated, audio-only voice-cloning detector that classifies short speech segments as bonafide or spoofed using a Wav2Vec2-based classifier (augmented with lightweight prosodic features) trained on ASVspoof2019 LA, fuses in a speaker-similarity signal from a lightweight enrollment check, aggregates segment scores into a calibrated Low/Medium/High risk output with a rule-based explanation and a recommended action (call-back, MFA, escalation), and

reports — rather than claims to solve — its accuracy drop when cross-tested on an Indian-speaker dataset (IndieFake). Raw audio is discarded after inference, with only features and scores retained. The system is delivered as a working demo (upload or record a clip, receive a risk report) behind a documented REST API, not a production telephony service.

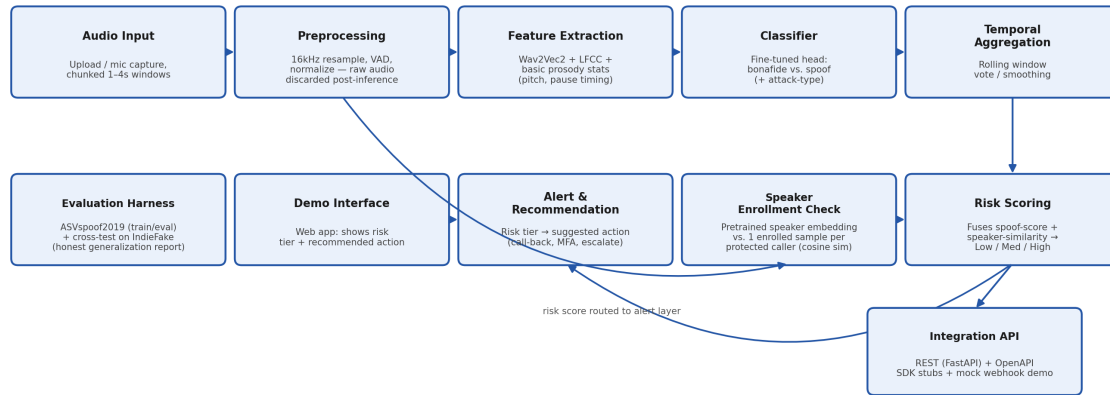
The pipeline is exposed behind a documented REST API (not gRPC — that's deferred) with an OpenAPI spec and generated SDK stubs, plus one mock webhook demo showing how a contact-center or communication tool would call it. This satisfies the "platform and integration" requirement in the problem-statement description with something real and demoable, without pretending to have live hookups into an actual bank, telecom carrier, or named contact-center platform.

What makes this defensible in front of judges is precisely that it does not overreach: every claim in the demo — the accuracy numbers, the "real-time" framing, the risk score, the speaker-similarity check, the API — corresponds to something actually running, and every gap the team cannot close in the build window (streaming stability, telephony robustness, research-grade privacy-preserving features, broad Indic coverage, fairness auditing, codec-specific attacks, multimodal checks, gRPC, real system integrations, live contextual enrichment, configurable workflows, real alert dispatch, formal compliance certification) is named as future work rather than glossed over.

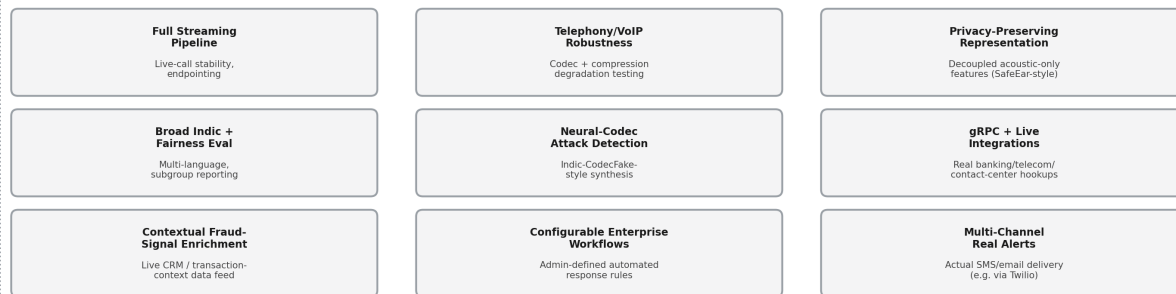
6. System Architecture

Blue boxes are built and demoed for the submission. The dotted panel below is explicitly future work — it should appear on the architecture slide too, so judges see the boundary rather than infer it.

Proposed System Architecture — SIH25104 (Buildable Scope, full PS coverage)



Acknowledged as future work (not in hackathon build)



6.1 Component breakdown

Component	What it does	How it's built
Audio Input	Accepts an uploaded file or microphone recording; splits it into overlapping 1–4 second windows	Browser mic capture / file upload; server-side chunking
Preprocessing	Resamples to 16kHz, trims silence with voice-activity detection, normalizes amplitude; discards the raw audio buffer immediately after inference	torchaudio / librosa; a standard VAD (e.g. webrtcvad)
Feature Extraction	Converts each window into a fixed-size representation, combining spectral and prosodic cues	Pretrained Wav2Vec2 (frozen or lightly fine-tuned) + LFCC baseline, plus lightweight prosodic statistics (pitch-contour variance, pause timing) via librosa/parselmouth
Classifier	Scores each window as bonafide or spoofed (attack-type sub-classification as a stretch goal)	Small classifier head (logistic regression / shallow MLP / XGBoost) on top of pooled embeddings, trained on ASVspoof2019 LA

Component	What it does	How it's built
Speaker Enrollment Check	Compares the incoming caller's voice against one enrolled reference sample for a protected caller (e.g. a CXO), producing a similarity score	Pretrained speaker-embedding model (e.g. SpeechBrain ECAPA-TDNN) + cosine similarity — no training required
Temporal Aggregation	Combines window-level scores into one utterance-level score	Rolling average / majority vote across windows — a simplified stand-in for full streaming stability logic
Risk Scoring	Fuses the spoof-detection score and speaker-similarity score into a calibrated, judge-legible output	Threshold-based Low / Medium / High buckets plus the raw confidence number
Explanation Tags	Attaches a short human-readable reason to the score	Rule-based mapping from which feature stream (spectral vs. phase) drove the decision
Alert & Recommendation	Maps the risk tier to a suggested action, shown as a pre-transaction warning	Rule-based lookup: Low → proceed, Medium → call-back/MFA, High → escalate to supervisor
Demo Interface	Lets a user try the system end-to-end	Simple web app (e.g. Streamlit/Flask + a small front end) showing the risk report and recommended action
Evaluation Harness	Produces the honest generalization numbers used in the report	Offline script: train/eval on ASVspoof2019, cross-test on IndieFake, log EER/accuracy for both
Integration API	Lets external systems (a contact center, a comms tool) call the detector and get a risk report back	FastAPI REST endpoint with auto-generated OpenAPI spec; SDK stubs generated via openapi-generator (Python + JS); one mock webhook script demonstrating the integration pattern end-to-end

7. Evaluation Plan

- Primary training/eval: ASVspoof2019 LA — report Equal Error Rate (EER) and accuracy on the standard eval split.
- Generalization check: cross-test the frozen model on IndieFake (Indian-speaker English speech) — report EER/accuracy without retraining, alongside the in-domain numbers, so the gap is visible rather than hidden.
- Latency: measure per-window inference time to support the "real-time-simulated" claim (target: comfortably under the window length, e.g. sub-second per 1–4s chunk).
- Ablation (if time permits): Wav2Vec2 front-end vs. classical LFCC front-end, to justify the feature-extraction choice with a number rather than a citation alone.
- Speaker enrollment check: report impostor-rejection and genuine-acceptance rates on a small set of enrolled test speakers, using cosine-similarity thresholds.
- Privacy check: confirm by log audit that no raw audio persists after inference — only features and risk scores are written to storage.

8. One-Line Framing for the Pitch

"We built and evaluated a working voice-cloning risk-scoring pipeline on public benchmarks, honestly measured how it degrades on Indian-speaker data, and scoped the remaining gaps — streaming robustness, telephony conditions, privacy-preserving detection, broader Indic coverage — as a clear next-phase roadmap."